

Build and Operate Foundry Agents

Build, deploy, govern, evaluate, and operate a cumulative Healthcare Marketplace Concierge using Microsoft Foundry Hosted Agents and Microsoft Agent Framework.

DURATION**4-hour core + optional stretch****DELIVERY****Instructor-led, hands-on****LEVEL****Developer • Python / code-first****WORKSHOP DESCRIPTION**

Software engineers progress from a local Agent Framework application to versioned hosted agents running in Microsoft Foundry. One synthetic healthcare marketplace scenario connects typed tools, governed knowledge, resilient sessions, multi-agent human handoff, observability, evaluations, release gates, and safe promotion practices.

OBJECTIVES AND EXPECTED OUTCOMES**Build the hosted product**

Create an Agent Framework agent with typed tools, Entra identity, shared compliance instructions, and the Responses protocol.

Ground and retain context

Add a Foundry IQ knowledge base over MCP and externalize conversation history for process restart and scale-out resiliency.

Orchestrate governed work

Implement specialist fan-out/fan-in, bounded compliance reflection, structured handoff packets, and advisor approval across HTTP turns.

Operate with confidence

Trace the shipped endpoint, evaluate quality and policy, enforce a fail-closed release gate, and practice version promotion and rollback.

Key delivery decision — notebook = cockpit; hosted/main.py = product. Learners build, test, deploy, invoke, and inspect from the dev container. Foundry builds and runs the same flat Python package as a managed, versioned container. Deployment is always an explicit reviewed action; nothing deploys during bootstrap or offline validation.

TIMED AGENDA

TIME	MODULE	OUTCOME
0:00–1:00	1. Hosted agent basics	Run locally, deploy from source with azd, invoke the active Responses version, and inspect status and logs.
1:00–2:00	2. Knowledge + sessions	Create Foundry IQ knowledge over MCP; kill and restart the agent while preserving conversation history.
2:00–3:00	3. Multi-agent handoff	Route across specialists, apply a compliance gate, and pause for licensed-advisor approval across turns.
3:00–4:00	4. Operate hosted agents	Trace, evaluate, gate, version, promote, smoke-test, and roll back without stored deployment secrets.
Optional	Stretch 5 + 6	Compare platform-managed workflows with hosted code; explore Invocations, bundled Skills, and optional Toolbox.

Prepare the room. Build cumulatively. Operate safely.

INTENDED PARTICIPANTS

Software engineers who use Python, VS Code, Git, notebooks, and command-line tools. The delivery is code-first and assumes comfort reading and modifying runnable examples.

REQUIRED READINESS

Workstation VS Code, Docker Desktop with Linux containers, Dev Containers extension, Git, and a Bash terminal.	Supported alternative GitHub Codespaces when organizational policy and Azure network access permit it.
Azure / Foundry Foundry account and project, gpt-5.4-mini-class chat deployment, text-embedding-3-large, Azure AI Search with Foundry IQ, and Application Insights.	Identity + permissions Entra sign-in; Azure AI User/Developer, Search contributor roles, Monitoring Reader, and invoke rights. Foundry Project Manager is required only for deployers.

TOOLS AND ENVIRONMENT

- Repository Python 3.14 dev container and pinned root requirements.
- Azure CLI, Azure Developer CLI, and the `azure.ai.agents` azd extension.
- Repository-root `.env`, unique attendee resource suffix, and documented project/search endpoints.
- Optional: existing Azure Blob container for shared Lab 2 history; local Azurite is available for emulator testing.

Safety and governance boundary. All workshop data is synthetic. Agents must verify identity, minimize and redact sensitive data, cite governed knowledge, never recommend or rank plans, never enroll a participant, and route judgment to a licensed advisor. Authentication is Entra-only; no keys or hardcoded endpoints. Do not grant roles, change networking, deploy, promote, or delete resources without approval.

CUMULATIVE LAB SEQUENCE

1	60 min	Hosted agent basics Agent + typed tools + compliance block + local Responses host + explicit azd deployment.
2	60 min	Knowledge and durable sessions Foundry IQ over MCP, external history, and a kill/restart continuity proof.
3	60 min	Multi-agent triage and handoff Explicit workflow graph, bounded review, structured packet, and human approval.
4	60 min	Operate hosted agents OpenTelemetry, deterministic policy evaluators, release gate, versions, promotion, and rollback.
S5	60 min	Prompt agents and workflows Optional comparison of platform-managed definitions and hosted-agent delegation.
S6	45–60	Invocations, Skills, and Toolbox Optional structured batch protocol, bundled skill, and preview Toolbox integration.

Delivery methodology
 Each core breakout follows **teach 10 • demo 10 • do 35 • checkpoint 5**. Learners use a generated notebook or its adjacent Python driver, produce an artifact consumed by the next lab, and can use `catch_up.py` to restore the cumulative chain.

CONSEQUENTIAL ACTIONS

Local build, smoke tests, and offline validation are required. Individual Azure deployment may be replaced by a proctor deployment when learners lack Foundry Project Manager. Cloud promotion, the nested GitHub Actions template, Azure Blob, Foundry evaluations, Toolbox, and preview workflow surfaces are optional and must be explicitly configured.

Included Four core labs, two optional stretch labs, synthetic data, notebooks and Python drivers, hosted packages, guardrails, evaluation data, offline validation, catch-up tooling, and dry-run-first cleanup.	Out of scope Production APIs or customer data, custom MCP servers, Teams/M365 publishing, a web-chat front end, A2A, Routines, Memory Store, Model Router, and automatic infrastructure provisioning.	Offering notes Use case: Healthcare Marketplace Concierge. Alignment: Develop AI agents on Azure (AI-3026). Source: build-and-operate-foundry-agents/. Updated 2026-10-01 • Reconciled from the current workshop, setup, use-case, sequence, and lab guides.
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